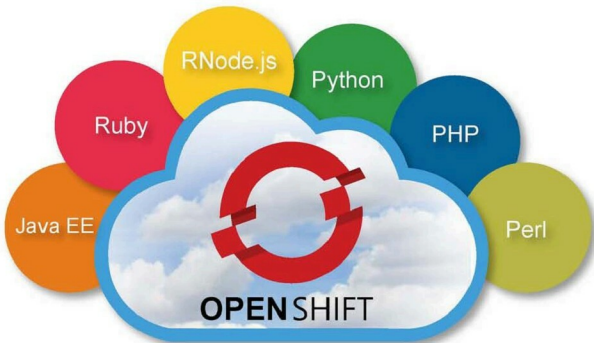


OpenShift: The New Age Platform-as-a-Service

A first of its kind, OpenShift is a cloud-based PaaS offering with complete support for Java EE and other languages like Ruby, Node.js, Python, PHP and Perl. Red Hat has just started offering the enterprise solution, with an obvious focus on easy scalability.



Cloud computing has often been perceived by old and new developers as full of unnecessary jargon. Terms like IaaS, PaaS and the ever-so-confusing products like Amazon EC2, Google App Engine, OpenStack and now OpenShift by Red Hat make it all the more difficult to understand where to slot each one of them in the application stack. Then there are different pricing models and APIs that need to be understood before actually diving into application design.

OpenShift has been a recent entrant into this field, and is being promoted heavily by Red Hat. The enterprise version of this product was recently launched, which has led to a lot of hype among developers. It is actually a Platform-as-a-Service solution, which allows you to create, deploy and manage applications on the cloud. These work on a set of 'gears', or more precisely, containers. The source code of the OpenShift project has

been released under the name OpenShift Origin, and it supports a lot of languages and application frameworks. There are two ways to interact with OpenShift, i.e., the command-line tool and the online control panel. There is also IDE integration provided by JBoss tools that can be used to directly deploy applications onto the cloud and manage them from the development workspace.

Let's take a look at the architecture.

Gears and nodes

OpenShift works on a set of 'gears' that act like containers with finite resources, and each application layer can be run on such a gear. The gears are available in multiple sizes, i.e., small and medium, with 512 MB and 1 GB of RAM. These gears work on machines or servers on which resources like memory and disk are constrained according to each gear. So, each node can have more than one gear running on them.